

conventional blockchain networks into their daily processes because they are publicly accessible. For obvious reasons, businesses don't want to share sensitive information or mission-critical business contracts. Hyperledger Fabric has been created to address this challenge. It is a high security foundation to facilitate enterprise blockchain solutions. It can be scaled and isolated to suit performance and privacy requirements of businesses.

Hyperledger Fabric allows verification of identities based on access permissions and enables membership authorisation or restriction, preventing unauthorised entry into the network by external entities. It also has strong technology for enabling companies to comply with government regulations on data protection like storing participants' data and their detailed access history.

A few exclusive benefits of Hyperledger Fabric are briefly described below.

Modular architecture

The Hyperledger Fabric module design divides the transaction processing workflow into three modules: chaincode, transaction ordering, and transaction validation and commitment.

Hyperledger cleans up the network and allows simplified, smooth operations by lowering trust levels and simplifying verification. It also aids in boosting the network's scalability and overall performance.

For example, once a rule based access module has been established, the company can reuse it for numerous identical cases, saving the time and effort needed to build the same module for each one.

The network participants assume three distinct roles:

- Endorser
- Committer
- Consenter

The number of endorsements required for a transaction proposal follows a predetermined guideline. As a result, this proposal gets sent to endorsing entities or peers. After receiving sufficient endorsements, the committers are handed the transaction block/batch to officially check that the standard endorsement procedure has been followed and that no different transactions have been discovered. The transactions are finally committed to the ledger after these two tests.

Only confirming instructions, such as read/write sets and signatures, are sent across the network to prevent latency while boosting performance and scalability. And only a few selected individuals have access to essential data in order to achieve and maintain maximum security. This also inhibits data theft/manipulation by unauthorised/external organisations because transactions may only be seen by committers and endorsers.

Example: A confectionery business in Spain wishes to sell Spanish turrón in the country at a lower price than in other nations. Its goal is to keep these prices hidden from other marketeers, effectively preventing them from discovering the prices for marketplaces in Spain.

However, this is not easy while following traditional blockchain technology because product logistics require a variety of parties such as shipping firms, financing institutions, customs officials, and others. The pricing can be easily found thanks to the distributed ledger and interconnected blocks.

Hyperledger facilitates the separation of private and public information, allowing sensitive data (in this case, prices) to be restricted solely to the proprietary network

and authorised entities, while other important information is shared on a public network.

Data rules compliance

Any legal entity operating in the digital sphere must comply with data protection regulations, and failing to do so could land it in legal trouble. Compliance is even more critical in some sensitive use cases/sectors, such as healthcare or financial services. It makes precise information about each network participant with access to specialised data obligatory for organisations. Hyperledger Fabric provides a dependable structure for enabling permission networks, i.e., each network participant has a known identity, allowing for easy tracing.

A private equity firm that is not publicly traded on the stock exchange is an example of a legal entity. Because it involves significantly larger sums of money, it's important to understand the financial credibility of the investors, who are typically angel investors, venture capital firms, or other private equity firms. The Hyperledger Fabric design allows only known and eligible players to participate in the blockchain.

Improving supply chain management

Apart from meeting compliance requirements, Hyperledger Fabric also aids in the reduction of counterfeiting and theft by allowing businesses to track conflict goods across the supply chain. It protects the interests of transporters, buyers and suppliers, while maintaining integrity and discouraging counterfeiting. Scalability, performance and trust levels are all improved.

One of the difficulties with the blockchain ecosystem is the scattered trust and verification points among different nodes,